

Time: Three Hours

Maximum Marks: 75

SECTION– A ($10 \times 2 = 20$ marks)**Answer any TEN questions**

- Write down the general term of Binomial theorem.
- Write down the formula for exponential series.
- State logarithmic series.
- Define Hermitian matrix.
- Write down the condition for orthogonal matrix.
- Define unitary matrix.
- Write down Lagrange's interpolation formula.
- Write down the Newton's backward interpolation formula.
- State DE Moivre's theorem.
- Write down the formula for $\tanh 3x$.
- Find $\frac{dy}{dx}$ if $y = ae^{mx} + be^{-mx}$.
- If $f(x) = \cos(ax + b)$ then find $D^n[f(x)]$.

SECTION– B ($5 \times 5 = 25$ marks)**Answer any FIVE questions**

- Find the coefficient of x^n in $(3+2x)e^{3x}$.
- If $A = \begin{bmatrix} 3 & 1 & -1 \\ 0 & 1 & 2 \end{bmatrix}$ then find $A^T A = AA^T$.
- Apply Lagrange's formula to find $f(5)$, given that $f(1) = 2, f(2) = 4, f(3) = 8, f(7) = 128$.

16. Using Newton's forward interpolation formula, find $f(1.02)$ from the following table

x	1.0	1.1	1.2	1.3	1.4
f(x)	0.841	0.891	0.932	0.964	0.985

- Express $\cos 6\theta$ in terms of $\cos\theta$.
- Prove that $\sin^6\theta = \frac{1}{32}[\cos 6\theta - 6\cos 4\theta + 15\cos 2\theta - 10]$
- If $x = \cos h\phi \cos\theta, y = a \sin h\phi \sin\theta$ then show that $\frac{\partial(x,y)}{\partial(\phi,\theta)} = \frac{a^2}{2}[\cosh 2\phi - \cos 2\theta]$

SECTION– C ($3 \times 10 = 30$ marks)**Answer any THREE questions**

- Prove that $\frac{a-x}{a} + \frac{1}{2}\left(\frac{a-x}{a}\right)^2 + \dots = \log a - \log x$.
- Verify Cayley- Hamilton theorem for $A = \begin{pmatrix} 1 & 0 & 3 \\ 2 & 1 & -1 \\ 1 & -1 & 1 \end{pmatrix}$ hence find its inverse.
- Find by Newton's method, the real positive root of $x = \cos x$, correct to 3 decimal places.
- If $u + iv = \cos h(x + iy)$, prove that $\frac{u^2}{\cosh^2 x} + \frac{v^2}{\sin^2 h^2 x} = 1$ and $\frac{u^2}{\cos^2 y} - \frac{v^2}{\sin^2 y} = 1$.
- If $y = \sin^{-1}x$ prove that $(1 - x^2)y_{n+2} - (2n + 1)xy_{n+1} - n^2y_n = 0$.